

a ladder activity

Equations of Parallel and Perpendicular Lines

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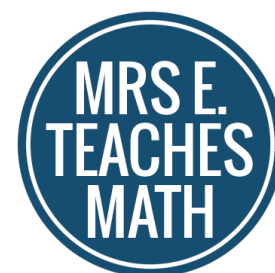
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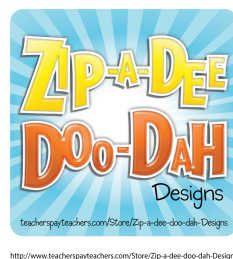
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If you have any questions or comments please email me at mrseteachesmath@gmail.com.

Thank you for respecting my work!



A special thank you to the following artists:



Instructions:

- The two following “worksheet” pages should be copied single-sided and given to each student or group of students. I like to have students work in partners for this activity.
- Students will work the problems in the space provided.
- When they are finished, students will cut along the lines to make a total of 12 squares.
- The first card says “Start”. Students will then match that card to the answer that is on the top of another card. They will continue in this fashion, making a chain. They will tape their completed chain together.

Suggestions:

- I like to copy the problems on colored paper and then hang a few of the completed chains in my classroom as decoration.
- These cards could be copied on cardstock and laminated for reuse.
- If you're concerned about the amount of paper, this could be copied two per page. The squares will just be smaller.
- Solutions are included at the end of the document.

Start

Write an equation of the line with y-intercept 3 that is parallel to $y = -\frac{3}{4}x - 4$.

$$y = -3x + 5$$

Write an equation of the line with y-intercept -2 that is perpendicular to $y = 4x + 6$.

$$y = -\frac{3}{4}x + 3$$

Write the equation of a line that is perpendicular to the line $3x + 2y = 8$ and passes through the y-intercept of that line.

$$y = \frac{1}{2}x - 2$$

Write the equation for the line that passes through $(4, 7)$ and is perpendicular to the line $y = \frac{2}{3}x - 1$.

$$y = -\frac{5}{3}x + 1$$

Write an equation of the line that is parallel to $y = -3x + 2$ and passes through the point $(2, 1)$.

$$y = -\frac{1}{4}x - 2$$

Line j passes through the points $(2, 1)$ and $(0, 0)$. Line k passes through the point $(4, 3)$ and is parallel to line j . Find an equation for line k .

$$y = 7x$$

End

$$y = \frac{2}{3}x + 4$$

Line j passes through the points $(-5, -2)$ and $(5, 4)$. Line k passes through the point $(-3, 6)$ and is perpendicular to line j . Find an equation for line k .

$$y = -\frac{3}{2}x + 13$$

Write the equation of a line that is perpendicular to the line $5x - 3y = 12$ and passes through the point $(3, 1)$.

$$y = \frac{1}{2}x + 1$$

Write the equation for the line that is parallel to the line $y = 7x - 3$ and passes through the origin.

$$y = -3x + 7$$

Write an equation of the line that passes through the point $(4, 0)$ and is perpendicular to $y = -2x + 1$.

$$y = -\frac{3}{5}x + \frac{14}{5}$$

Line j passes through the points $(-4, 10)$ and $(2, -8)$. Line k passes through the point $(1, 2)$ and is parallel to line j . Find an equation for line k .

Writing Equations of Lines Chain Solutions

1. Start

2. $y = -\frac{3}{4}x + 3$

3. $y = \frac{2}{3}x + 4$

4. $y = -\frac{5}{3}x + 1$

5. $y = -3x + 7$

6. $y = \frac{1}{2}x - 2$

7. $y = -\frac{3}{2}x + 13$

8. $y = -\frac{3}{5}x + \frac{14}{5}$

9. $y = -3x + 5$

10. $y = -\frac{1}{4}x - 2$

11. $y = \frac{1}{2}x + 1$

12. $y = 7x$